

- 8 (a) (i) Similarity: B1
Both nuclear reactions produce energy
or
Both produces products that are more stable/higher binding energy per nucleon.
- Difference: B1
Nuclear fusion combines two lighter nuclei into a heavier nuclei while nuclear fission splits a heavier nuclei into two lighter nuclei of similar sizes.
- Nuclear fission produces radioactive waste while nuclear fusion does not.
- (ii) The process is able to use the energy produced to keep itself going without any external help/guidance. B1
- (iii) It does not produce any radioactive waste/ produces very little short-lived radioactive wastes. B1
- It produces more energy per unit mass compared to nuclear fission.
- Nuclear fusion reactors will not melt down compared to nuclear fission reactors.
- The raw ingredients required for fusion is more readily available compared than that for fission.
- (b) (i) High amount of kinetic energy required for the positively charged nuclei to B1
overcome the Coulombic/electrostatic repulsion B1
- (ii) To increase the probability of collision between the nuclei B1
- (c) (i) $4 m_H - (m_{He} + 2 m_e)$
 $= 4(1.007825) - (4.002604 + 2(0.000549))u$ M1
 $= 4.58(1) \times 10^{-29} \text{ kg.}$ A1
- (ii) $E = \Delta mc^2$
 $= (4.581 \times 10^{-29})(3.00 \times 10^8)^2$
 $= 4.123 \times 10^{-12} \text{ J } (/ 1.60 \times 10^{-19})$
 $= 25.8 \text{ MeV}$ A1

- (iii) Percentage loss in mass

$$= (4 \times 1.007825 - (4.002604 + 2(0.000549)) / (4 \times 1.007825)$$

$$= 0.006846$$

$$= 0.6846\%$$
 A1
- (iv) Loss in mass $M = M_{core} \times 0.006846 = 0.10 \times M_{Sun} \times 0.006846$ C1

$$3.8 \times 10^{26} = \frac{(0.10 \times M_{Sun} \times 0.006846)(3.00 \times 10^8)^2}{t}$$
 C1

$$t = 3.243 \times 10^{17} \text{ s}$$
 A1

$$= 10.3 \text{ billion years}$$
- (d) (i)
 - Inverted asymmetrical U-shaped curve with right side lower in height. B1
 - Max B.E per nucleon is between 8.0 to 9.5 MeV B1
 - Corresponding nucleon number is between 55 - 65 B1
- (ii) Iron has a very high binding energy per nucleon. M1
 When iron is fused, the products are less stable/lower BE per nucleon A1
 and there is no net energy released in the reaction.
- (iii) They are produced via explosions which provide energies for these B1
 elements to be fused together.